
Frozen VLA teachers
Benign prompt τ_b:
"put the bowl on the stove"
Target prompt τ_t:
"put the bowl on plate"
Frozen VLA model
A_b(o)
A_t(o)
Teacher actions
Observation (o)
Constraint aware candidate search
Near-benign candidate prompts (derived from τ_b)
τ1: "put the bowl on the staove"
τ2: "put the bowl on the st6ave"
τ3: "put the bol on the stave"
τn: "put the bol on the stave"
Filter using Eq. (2)
Rank using Eq. (8)
Closed-loop rollout selection
Evaluate M candidates in a fixed episode using Eq. (10)
T_e
B_e
Mean
Text
τ1
0.12
0.95
0.78
1.8
τ2
0.93
0.18
0.11
1.7
τM
0.67
0.61
0.24
1.5
Select the best adversarial prompt + useful near-miss / diverse rollouts
selected rollouts B
On-policy aggregation
For each selected prompt τ ∈ B, we store (o_k^τ, A_b(o_k^τ), A_t(o_k^τ), weight)
τi
A_b(o_k^τ)
A_t(o_k^τ)
Teacher labels for each observation